

Predicting Customer Churn for Targeted Retention in the Telecommunications Industry

Maggie Boles, Adina Kaplan, Logan McCorkle, & Chana Ochs

Bellevue University

DSC450-T301 Applied Data Science (2265-1) Spring 2026

Professor Fadi Alsaleem

04/26/2026

Predicting Customer Churn for Targeted Retention in the Telecommunications Industry

Domain

The focus of this project is the Telecommunications industry, specifically the economic challenge of customer churn. Because it is significantly more expensive to acquire a new customer than to retain an existing one, building an automated system to predict turnover is a high-priority strategic mission for modern service providers. Specifically, we are looking at subscription-based business models where long-term revenue depends on minimizing customer turnover.

Dataset Selection

We will use the IBM Sample Data Set for Telco Customer Churn, which includes 7,043 customer records and 21 distinct features. This dataset was chosen because it represents a realistic environment for a mid-sized service provider. It contains a mix of demographic data, service tiers (such as Fiber Optic and DSL), and financial billing cycles, which provides the necessary complexity to build a meaningful predictive model.

Research Questions

To evaluate the effectiveness of retention strategies within the telecommunications sector, this project seeks to answer the following questions:

1. Which specific service types or bundles are most likely to result in customer churn?
2. Is there a clear pricing threshold or specific billing behavior where the risk of a customer leaving increases significantly?
3. Which customer segments should be prioritized for retention offers to maximize the ROI of a marketing intervention?

Methodology

Data Engineering & Cleaning

Our methodology involves fixing data type errors, specifically converting the 'Total Charges' column from a string to a numeric format. We will handle missing values using median imputation to ensure dataset integrity. Because we are using both Random Forest and XGBoost, we will also verify that there are no conflicting data types that could cause gradient-boosting errors during training.

Feature Engineering

We will develop new metrics, such as "Tenure-to-Charge" ratios and service counts, to better understand usage patterns. Categorical data will be converted using One-Hot Encoding. This encoding strategy is essential for both the Random Forest and XGBoost algorithms to process categorical variables as distinct numerical inputs.

Modeling

We will implement a comparative modeling approach using Random Forest and XGBoost (Extreme Gradient Boosting) classifiers. Both models are ideal for identifying complex, non-linear patterns in customer behavior. The Random Forest will provide a robust baseline for ensemble learning, while XGBoost will be used to improve predictive accuracy through its iterative gradient-boosting framework. By using both, we can compare "Feature Importance" rankings from each model to verify which factors most consistently influence a customer's decision to churn.

Evaluation

We will assess and compare model performance using a Confusion Matrix, F1-Score, and AUC-ROC curves. This approach allows us to accurately identify customers who plan to

leave, rather than relying on overall accuracy, which can be misleading in imbalanced datasets. By prioritizing the F1-Score and AUC-ROC, we ensure both the Random Forest and XGBoost models are specifically penalized for false negatives—failing to identify customers who actually churn—which is the most costly error for the business. Comparing the AUC-ROC curves will further help us determine which model is more robust at distinguishing between churn and non-churn across various probability thresholds.

Anticipated Challenges

Class Imbalance

Since most customers do not churn, the data is naturally skewed toward retention. This imbalance can lead a model to prioritize accuracy by simply predicting that no one will leave, rendering it useless to the business. To mitigate this, we will implement the Synthetic Minority Over-sampling Technique (SMOTE) during the training phase. This ensures the model learns the specific characteristics of the minority class (churners) and improves its ability to accurately predict them.

Data Leakage

A significant risk in churn prediction is data leakage, where features are included that wouldn't be available at the time of prediction. For example, a "reason for cancellation" column cannot be used to predict whether someone will cancel. We will conduct a thorough feature audit to ensure that no information collected after a customer has initiated the churn process is included in the training data, as this would result in an unfairly biased and non-functional model.

Model Complexity and Overfitting

Using an XGBoost model increases the risk of overfitting due to the algorithm's high sensitivity to hyperparameter settings. To address this, we will utilize cross-validation and Grid

Search to optimize parameters such as the learning rate and max depth. This ensures the model generalizes well with new, unseen customer data rather than simply memorizing the training set.

Abstract

This project builds a predictive pipeline to identify customers at risk of leaving a telecommunications provider. Using a dataset of over 7,000 records, we will focus on cleaning "messy" billing data and engineering new features that reflect customer loyalty. By training a Random Forest Classifier, we aim to pinpoint the specific pain points, such as contract lengths or internet speeds, that cause turnover. The end goal is to provide a list of high-risk customers so the company can intervene with targeted retention offers, ultimately protecting the company's bottom line and increasing customer lifetime value.

References

- Burez, J., & Van den Poel, D. (2009). Handling class imbalance in customer churn prediction. *Expert Systems with Applications*.
- Gartner. (2024). *Definition of customer churn and retention metrics*. Gartner Glossary.
- Hadden, J., et al. (2007). Computer simulation of customer churn. *Journal of Operational Research Society*.
- Huang, B., et al. (2012). Customer churn prediction in telecommunications. *Expert Systems with Applications*.
- Kaggle. (2024). *Telco customer churn by IBM sample data sets*.
<https://www.kaggle.com/datasets/blastchar/telco-customer-churn>
- Lariviere, B., & Van den Poel, D. (2005). Predicting customer retention and profitability. *Journal of Service Research*.
- Stripe. (2024). *Economic impacts of churn on SaaS and subscription models*. Stripe Business Guides.
- UCI Machine Learning Repository. (2024). *Bank marketing dataset documentation*.
<https://archive.ics.uci.edu/ml/datasets/Bank+Marketing>
- Verbeke, W., et al. (2011). Social network analysis for customer churn prediction. *Applied Soft Computing*.
- Xue, P., et al. (2011). Customer churn prediction using improved decision trees. *Journal of Software*.